

Foundations for programming languages 5: Answers for exercises

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Exercise 1

Evaluate an arithmetic expression

$$((4 + Y) * (5 + Z))$$

under the state $\sigma = \{ (X, 3), (Y, 20), (Z, 13) \}$.

An answer

Evaluate an arithmetic expression

$$((4 + Y) * (5 + Z))$$

in the state $\sigma = \{ (X, 3), (Y, 20), (Z, 13) \}$.

$$\begin{array}{rcl} < 4, \sigma > \rightarrow 4 & < Y, \sigma > \rightarrow 20 & \\ \hline < (4 + Y), \sigma > \rightarrow 24 & & \\ & & < 5, \sigma > \rightarrow 5 & < Z, \sigma > \rightarrow 13 \\ \hline & & < (5 + Z), \sigma > \rightarrow 18 & \\ \hline & & < ((4 + Y) * (5 + Z)), \sigma > \rightarrow 432 & \end{array}$$

Exercise 2

Let $\sigma = \{ (X, 10), (Y, 20), (Z, 30) \}$.

Write down all the elements of $\sigma[40 / X]$ in the set notation.

An answer

Let $\sigma = \{ (X, 10), (Y, 20), (Z, 30) \}$.

Write down all the elements of $\sigma[40 / X]$ in the set notation.

$$\sigma[40 / X] = \{ (X, 40), (Y, 20), (Z, 30) \}$$

Exercise 3

Derive the state after executing the statement

$X = (Y + 2); Y = (Y + 3);$

in the state $\sigma = \{ (X, 10), (Y, 20), (Z, 30) \}$.

An answer

Derive the state after executing the statement

$X = (Y + 2); Y = (Y + 3);$

in the state $\sigma = \{ (X, 10), (Y, 20), (Z, 30) \}$.

$$\begin{array}{c}
 \frac{\frac{\langle Y, \sigma \rangle \rightarrow 20 \quad \langle 2, \sigma \rangle \rightarrow 2}{\langle (Y + 2), \sigma \rangle \rightarrow 22}}{\langle X = (Y + 2); \sigma \rangle \rightarrow \sigma[22/X]} \quad \frac{\frac{\langle Y, \sigma[22/X] \rangle \rightarrow 20 \quad \langle 3, \sigma[22/X] \rangle \rightarrow 3}{\langle (Y + 3);, \sigma[22/X] \rangle \rightarrow 23}}{\langle Y = (Y + 3); \sigma[22/X] \rangle \rightarrow (\sigma[22/X])[23/Y]} \\
 \hline
 \langle X = Y + 2; Y = Y + 3;, \sigma \rangle \rightarrow (\sigma[22/X])[23/Y] \\
 \parallel \\
 \{ (X, 22), (Y, 23), (Z, 30) \}
 \end{array}$$

Exercise 4

Derive the state after executing the statement

`while ( Y ) { Y = (Y - 20); }`

in the state $\sigma = \{ (X, 10), (Y, 40), (Z, 30) \}$.

An answer

An answer is given in another file due to the lack of space.